Motion Sensor-Based Keyword Inference in Smart Home Systems

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Abstract

This work investigates whether vibration signals captured using a motion sensor can be used to infer spoken keywords. Specifically, an ADXL345 accelerometer is used to record vibration data generated by audio playback, and a Convolutional Neural Network (CNN) is trained to classify these signals into predefined keyword categories. The model is first trained and evaluated on a structured accelerometer dataset under controlled conditions, achieving strong classification performance. To assess real-world applicability, the trained model is then tested on vibration data collected using a Raspberry Pi and external speaker setup.

Experimental results show a significant drop in performance when transitioning from dataset-based evaluation to real-world testing, highlighting the challenges of generalizing across different environments. More importantly, the findings demonstrate that motion sensors have the potential to act as unintended side channels for speech-related information. This raises important privacy and security concerns in IoT systems, where such sensors are widely accessible. Overall, this work provides both a proof of feasibility for vibration-based keyword inference and an initial exploration of its implications for privacy and real-world deployment.

1. Introduction

Voice-controlled systems have become a core component of modern IoT environments, enabling users to interact with devices through spoken commands. While these systems are designed around audio processing, they also produce unintended physical side effects, such as vibrations generated during sound playback. These vibrations propagate through surrounding surfaces and can be captured by nearby sensors.

This project investigates whether such vibration signals, captured using an accelerometer, can be used to infer spoken keywords. Specifically, an ADXL345 motion sensor is used [1] to record vibration data generated by audio playback from a speaker. The recorded signals are then processed and classified using a Convolutional Neural Network (CNN) [3, 6] to determine whether meaningful keyword information can be recovered. CNN-based models have been widely used for time-series and sensor based classification tasks [2, 10].

The study is conducted in two stages. First, a CNN model is trained and evaluated using a structured accelerometer dataset containing labeled keyword signals. Second, the trained model is tested on real-world vibration data collected using a Raspberry Pi and external speaker setup [8]. This allows for a direct comparison between controlled dataset performance and real-world deployment performance.

The primary objective of this work is to evaluate the feasibility of motion sensor-based keyword inference while investigating potential privacy risks, where motion sensors may unintentionally act as side channels for recovering speech related information. In addition, the study analyzes how well models trained on curated datasets generalize to real-world conditions.

These findings highlight important security and privacy implications, as motion sensors embedded in everyday devices may unintentionally leak sensitive information. Unlike microphones and cameras, IMU sensors are commonly classified as zero-permission sensors because applications can often access them without requesting explicit user authorization. This creates a potential security risk in which malicious applications may silently collect vibration signals in the background and infer speech-related information without user awareness.

The results also reveal a significant gap between dataset performance and real-world performance, emphasizing the challenges of deploying machine learning models outside controlled environments.

2. Related Work

Deep learning has been widely applied in areas such as speech recognition, signal processing, and sensor-based classification [3, 6]. Convolutional Neural Networks (CNNs), in particular, have demonstrated strong performance in extracting features from time-series data, including both audio and motion signals[2, 10].

Accelerometer-based systems have been extensively used for applications such as activity recognition, gesture detection, and human motion analysis. These approaches typically rely on structured datasets where signals are clean, consistent, and well-labeled. However, using accelerometer data to infer audio content or spoken words remains a relatively unexplored direction.

Recent research has shown that motion sensors can act as side channels capable of leaking sensitive information. Prior work demonstrates that accelerometer and gyroscope data can be used to infer keystrokes, user interactions, and even speech-related patterns by analyzing subtle physical signals generated during device usage. These findings highlight potential privacy risks, especially in IoT and mobile environments where motion sensors are widely accessible and often lack strict access control.

One important example is the StealthyIMU framework proposed by Sun et al. [9], which demonstrated that zero-permission IMU sensors can be exploited to infer speech-related information from smartphone voice assistants. Their work highlights the potential privacy risks associated with accelerometer and gyroscope sensors, even when microphones are protected by explicit user permissions.

In addition, prior studies have shown that models trained on controlled datasets often struggle when applied to real-world data due to variations in noise, signal quality, and environmental conditions. This challenge, commonly referred to as domain mismatch, remains an important consideration when deploying machine learning systems in practical settings.

This project builds on these research directions by investigating whether vibration signals captured from audio playback can be used for keyword classification, while also exploring their potential as a side channel for leaking speech-related information. The study further examines how model performance changes when transitioning from controlled datasets to real-world conditions.

3. Methodology

3.1. System Overview

The proposed system is designed to investigate whether vibration signals captured by a motion sensor can be used to infer spoken keywords. The system consists of four main components: an audio source, a speaker, a motion sensor, and a machine learning model.

Pre-recorded keyword audio files are played through an external speaker, generating vibrations that propagate through the surrounding environment. These vibrations are captured using an ADXL345 accelerometer connected to a Raspberry Pi. The recorded signals are then processed and used as input to a Convolutional Neural Network (CNN), which classifies the signals into predefined keyword categories.

The overall objective of the system is to establish a mapping between vibration patterns and spoken keywords, enabling keyword inference without directly accessing the audio signal.

3.2. Data Collection

Two types of data were used in this study: dataset-based accelerometer data and real-world vibration recordings.

The dataset-based data was obtained from a pre-existing IMU dataset containing labeled accelerometer recordings associated with spoken keywords. This dataset was used for training and initial evaluation of the model under controlled conditions.

In addition to the dataset, real-world data was collected using a Raspberry Pi [8] and an ADXL345 accelerometer [1]. During data collection, keyword audio samples were played through a speaker positioned near the sensor. The accelerometer recorded vibration signals at a sampling rate of 200 Hz over a duration of five seconds for each sample. A short baseline period of silence was included before each recording to capture background noise and stabilize the sensor.

Each recording was saved as a .acc file and organized into sample directories. A total of 100 real-world samples were collected to evaluate the model's performance under practical conditions.

Figure 1 illustrates a sample vibration trace captured by the ADXL345 accelerometer during keyword playback. The three axes (x, y, and z) represent the directional acceleration components, while the magnitude signal summarizes overall vibration intensity. Noticeable fluctuations occur during the audio playback interval, indicating that the sensor captures vibration patterns associated with the spoken keyword.

3.3. Data Preprocessing

The raw accelerometer data was preprocessed to ensure consistency and suitability for model training. Each recording contains three channels corresponding to the x, y, and z axes of the accelerometer (ax, ay, az).

To standardize the input format, all signals were resampled or truncated to a fixed length of 256 time steps. This ensures that all input samples have the same dimensionality, which is required for batch processing in neural networks.

Each channel was then normalized independently to reduce the impact of amplitude variations and improve numerical stability during training. The processed accelerometer signals were represented as a three-channel time-series tensor:

$$X \in \mathbb{R}^{3\times T} \tag{1}$$

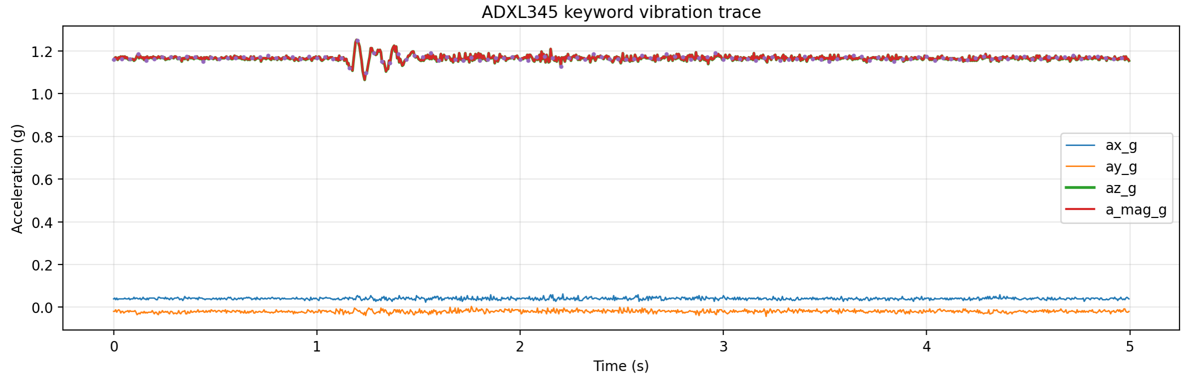

Figure 1. Example vibration trace captured by the ADXL345 sensor during keyword playback.

where the three rows correspond to the x, y, and z axes of the accelerometer, and T denotes the number of time steps. This preprocessing pipeline enables the model to focus on relative patterns in the vibration signals rather than absolute magnitude differences.

3.4. Model Architecture

A one-dimensional Convolutional Neural Network (1D CNN) was used to classify vibration signals into keyword categories. CNNs are well-suited for time-series data due to their ability to learn local temporal patterns [4] through convolutional filters.

The model consists of multiple convolutional layers that extract hierarchical features from the input signal. Each convolutional layer is followed by batch normalization and a ReLU activation function, which improves training stability and introduces non-linearity.

Max pooling layers are used to progressively reduce the temporal resolution of the feature maps, allowing the model to capture more abstract representations. Dropout layers are included to reduce overfitting by randomly disabling a portion of neurons during training.

The final layers of the network consist of fully connected layers that map the extracted features to output class probabilities. A softmax function is applied to produce a probability distribution over all keyword classes. The final layer applies a softmax function to convert the network outputs into class probabilities:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}} \tag{2}$$

where z_i represents the output score for class i, and C is the total number of classes.

The convolution operation used in the CNN extracts local temporal features from the input signal and can be expressed as:

$$(x * w)(t) = \sum_{k=0}^{K-1} x(t-k) \cdot w(k) \tag{3}$$

where x is the input signal, w is the convolution kernel, and K is the kernel size.

3.5. Training Procedure

The model was trained using supervised learning with labeled accelerometer data [3]. The objective was to minimize the cross-entropy loss between predicted and true labels.

The cross-entropy loss function used for training is defined as:

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{4}$$

where y_i is the ground truth label and $\hat{y}_i$ is the predicted probability for class i.

Training was performed using the AdamW optimizer [7], which combines adaptive learning rates with weight decay for improved generalization. A cosine annealing learning rate scheduler was applied to gradually reduce the learning rate over time, helping the model converge more effectively.

To address class imbalance in the dataset, class weights were incorporated into the loss function [5], ensuring that less frequent classes were given appropriate importance during training.

The model was trained for 30 epochs with a batch size of 64. During training, both training and validation accuracy were monitored to assess model performance and detect potential overfitting.

3.6. Evaluation Methodology

Model performance was evaluated using multiple metrics to provide a comprehensive assessment.

Overall accuracy was used as the primary metric to measure classification performance.

Overall classification accuracy is computed as:

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Samples}} \quad (5)$$

In addition, a confusion matrix was generated to analyze misclassification patterns between different keyword classes.

Per-label accuracy was computed to evaluate how well the model performs on individual keywords, highlighting variations in classification difficulty across classes. Prediction distribution analysis was also conducted to identify potential biases in the model's output.

Beyond dataset-based evaluation, the model was tested on real-world vibration recordings collected using the ADXL345 sensor [1]. This evaluation is critical for assessing the model's ability to generalize to practical deployment conditions.

4. Experimental Evaluation

4.1. Training Performance

The training behavior of the CNN model was evaluated by monitoring both training and validation accuracy across epochs. Figure 2 illustrates the learning curves obtained during training.

The model demonstrates a steady increase in both training and validation accuracy, indicating effective learning of underlying patterns in the accelerometer signals. The training accuracy gradually increases from approximately 35% in the early epochs to around 73% at convergence. Similarly, the validation accuracy improves consistently, reaching approximately 78%.

Notably, the validation accuracy is slightly higher than the training accuracy throughout training. This behavior can be attributed to the use of regularization techniques such as dropout and data augmentation, which make the training process more challenging while improving generalization. The convergence of both curves around epoch 25 indicates stable optimization without significant overfitting.

4.2. Dataset-Based Test Results

After training, the model was evaluated on a held-out test set derived from the same dataset. The model achieved an overall accuracy of approximately 80%, demonstrating strong performance under controlled conditions.

This result confirms that the CNN successfully learned meaningful features from the accelerometer data and can generalize well to unseen samples drawn from the same distribution. The high test accuracy, combined with stable validation performance, indicates that the model effectively captures patterns present in the dataset.

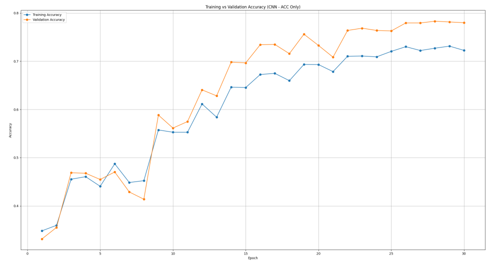

Figure 2. Training and validation accuracy across epochs for the CNN model.

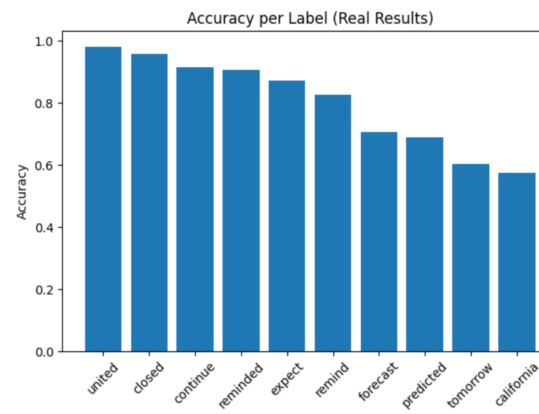

Figure 3. Per-label classification accuracy showing variation across different keywords.

4.3. Per-Label Performance Analysis

To further analyze model performance, accuracy was evaluated on a per-label basis. The results reveal significant variation across different keyword classes.

Certain keywords, such as united and closed, achieved very high accuracy (above 95%), indicating that their corresponding vibration patterns are highly distinguishable. In contrast, other keywords such as california and tomorrow exhibited lower accuracy (around 57-60%), suggesting that their vibration signatures are less distinct or more susceptible to noise.

This variation highlights that the discriminative power of vibration-based features depends strongly on the characteristics of the underlying audio signal.

4.4. Confusion Matrix Analysis

A confusion matrix was generated to analyze classification errors across all keyword classes. Figure 4 presents the confusion matrix for the test dataset.

The matrix reveals that while many classes are correctly classified, certain keywords are frequently misclassified into other classes. These misclassifications suggest that

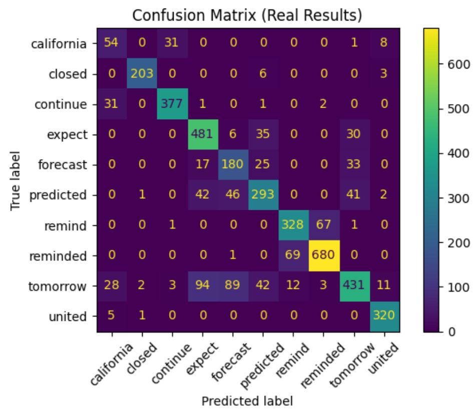

Figure 4. Confusion matrix illustrating classification performance across all keyword classes.

some vibration patterns share similar temporal characteristics, making them difficult to distinguish using accelerometer data alone.

In particular, keywords with lower per-label accuracy tend to exhibit higher confusion with multiple classes, reinforcing the observation that vibration-based signals may lack sufficient discriminative detail for certain words.

In the real-world evaluation, many predictions were concentrated around the "remind/reminded" related classes, indicating that the model developed a bias toward dominant or structurally similar vibration patterns under noisy conditions. This observation highlights the challenges of distinguishing subtle keyword differences in practical environments.

4.5. Prediction Distribution Analysis

To evaluate potential model bias, the distribution of predicted labels was analyzed. Figure 4 shows the frequency of predictions across all classes.

The results indicate that the model does not distribute predictions uniformly across all labels. Instead, certain classes are predicted more frequently, suggesting that the model may favor specific patterns that are more dominant in the training data.

This imbalance may be influenced by variations in class representation, signal clarity, or learned feature dominance.

4.6. Real-World Evaluation

To assess real-world applicability, the trained model was evaluated using vibration data collected from a physical setup consisting of a speaker and an ADXL345 accelerometer.

A total of 100 samples were recorded by playing keyword audio through a speaker and capturing the resulting vibrations. The model achieved an overall accuracy of approximately 12% on this real-world dataset.

This result represents a substantial decrease compared to the dataset-based accuracy, indicating that the model struggles to generalize beyond the conditions represented in the training data. This highlights the challenge of deploying models trained on controlled datasets in real-world environments.

Although the real-world accuracy is relatively low, this result remains an important finding because it reveals the limitations of current vibration-based inference models under practical deployment conditions. These findings emphasize the need for improved robustness, stronger generalization techniques, and more diverse real-world training data in future research.

4.7. Domain Mismatch Analysis

Figure 5 presents a direct comparison between dataset-based performance and real-world performance.

The model achieves approximately 80% accuracy on the dataset but only 12% accuracy on real-world vibration recordings. This performance gap cannot be explained solely by domain mismatch. A major contributing factor is the difference in data collection configurations between the open-source dataset and the real-world experimental setup used in this project. Variations in sensor placement, hardware configuration, environmental conditions, vibration propagation, and signal quality introduce substantial differences between the two data sources, making generalization significantly more difficult.

Additional contributing factors include differences in signal quality, environmental noise, sensor placement, and the propagation of audio-induced surface vibrations. These factors introduce variability that is not captured in the original dataset, leading to reduced model effectiveness. These findings demonstrate the importance of robust real-world evaluation when assessing vibration-based side-channel inference systems.

5. Conclusion

This work investigated whether vibration signals captured using an accelerometer can be used to infer spoken keywords. The results show that a CNN-based model can successfully learn meaningful patterns from structured accelerometer datasets and achieve strong classification performance under controlled conditions.

More importantly, these findings demonstrate that motion sensors have the potential to act as unintended side channels for speech-related information. This raises important privacy and security concerns in IoT systems, where motion sensors are widely accessible and often lack strict access control mechanisms.

At the same time, a significant gap between dataset-based performance and real-world performance was observed. This highlights the challenges of applying machine

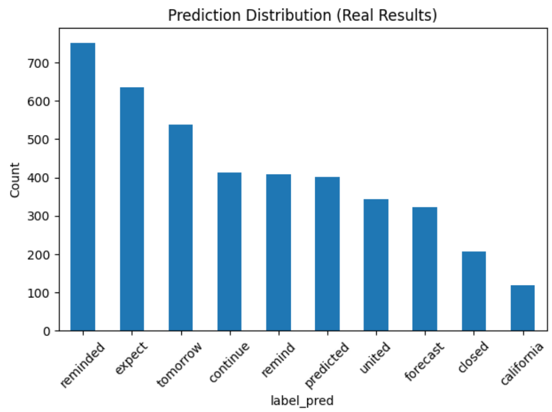

Figure 5. Distribution of predicted labels across the dataset, showing potential model bias.

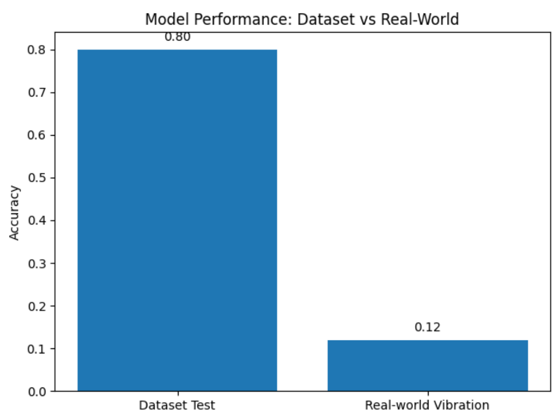

Figure 6. Comparison of model accuracy on dataset-based and real-world vibration data.

learning models outside controlled environments, where noise, environmental factors, and signal variability can degrade performance.

Overall, this work provides both a proof-of-concept demonstration of vibration-based keyword inference and an initial exploration of its implications for privacy, emphasizing the need for further research in more realistic and diverse conditions.

Future work will focus on improving robustness against environmental noise and hardware variability, expanding real-world data collection, and exploring more advanced deep learning architectures such as LSTM and Transformer-based models. Future research may also investigate domain adaptation techniques to improve transferability between controlled datasets and real-world sensing environments.

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